
10" MAXI Controller Pro User Manual

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1. 10" Maxi Controller Pro overview

Thank you for choosing 10" Maxi Controller Pro. This user manual provides comprehensive guidance for utilizing our high-end controller designed by Mobilicom. Specifically crafted to seamlessly integrate with the 10" Panasonic Toughbook, the 10" Maxi Controller Pro offers a wide range of configurable controls to enhance your operational experience. In this manual, you will find detailed instructions on setup, operation, and maintenance, ensuring optimal performance of your 10" Maxi Controller Pro.

The 10" Maxi Controller Pro is an "All in One" ruggedized Tactical Mobile Controller/GCS (Ground Control Station), meticulously designed to support drones and robotics operations with a variety of payloads. It offers various controlling options to support mission-critical operations, serving commercial and governmental sectors for uncrewed applications. This versatile controller is equipped to handle complex tasks with precision and ease, making it an indispensable tool for professionals in a wide range of industries. The 10" Maxi Controller Pro stands out for its seamless integration with Mobilicom's cutting-edge Software Defined Radio (SDR), ensuring reliable communication and superior performance in every operation. Throughout this manual, you will discover the full capabilities of the 10" Maxi Controller Pro, empowering you to maximize productivity and efficiency in your operations.



Figure 1: 10" Maxi Controller Pro

2. Main Features

The 10" Maxi Controller Pro offers the following features:

1. Customizable Control Interface:

Per user preferences, the 10" Maxi Controller Pro includes a control interface comprising configurable joysticks, buttons, toggles, and sliders. This customization empowers users with precise and intuitive control, crucial for navigating complex operational environments.

2. Ruggedized Durability:

Engineered to withstand harsh conditions, the 10" Maxi Controller Pro has a ruggedized design that ensures durability and reliability in challenging environments. Built to endure extreme conditions, this controller is well-suited for deployments in demanding operational landscapes, providing endurance and resilience in the field.

3. Unified All-in-One Design:

Featuring an integrated Toughpad multi-touch Tablet, the 10" Maxi Controller Pro offers a streamlined design. The 10" Maxi Controller Pro has seamless integration with Mobilicom's cutting-edge Software Defined Radio (SDR), ensuring reliable communication and superior performance in every operation. This unified layout enhances portability and operational efficiency, consolidating multiple elements into a single, user-friendly device.

4. Versatile Connectivity Options:

Offering flexible connectivity options, the 10" Maxi Controller Pro allows seamless integration with Mobilicom's advanced communication units or compatible third-party devices. This versatility enables users to adapt the controller to various communication setups, optimizing performance across diverse operational scenarios.

5. Real-time HD Video and Broadband Data Transmission:

The 10" Maxi Controller Pro enables seamless real-time HD video streaming and broadband data transmission, facilitating mission-critical operations where precision and instantaneous decision-making are essential.

3. Product Highlights

- Light & Robust
- Configurable multiple joysticks, buttons, sliders, toggle, switches
- Dust and Water-Resistant Design
- Daylight readable (1000 Nits)
- Includes 10" Panasonic Toughpad Tablet
- Seamless integration with Mobilicom's advanced Software Defined Radio (SDR)
- Compatibility with various communication setups for versatile deployment options
- User-friendly interface with intuitive navigation controls
- I/O interfaces to peripherals

4. Technical Specifications

Tablet Specifications	
Type	10" Toughpad FZ-G2 Panasonic
CPU	Intel core I5
User interface	Windows 10 Professional
Display	10" WXGA 1920 x 1200

Dimensions (H x W x D)	
24X43X3.5 Cm	9.4X16.9 X 1.3 Inch
Weight	
3.13 Kg	6.9 Pound
Power	
12-19V DC	Battery Li-ion
Environmental	
Temperature	Operating -20 to 50°C Storage -50 to 85°C
Humidity	5 to 95% non-condensing
Casing	Dust and water resistant
Interfaces	
External Interface	USB 2.0, Ethernet
Human Interface	2 joysticks, 6 buttons, 3 sliders, 2 toggle Switches. User configurable
Antenna Connector	3 or 5 x SMA: 2 x RF, 1 x GPS, 2 x LTE - optional

Table 1: Technical Specifications

5. Control Interface Overview

The 10" Maxi Controller Pro offers extensive configuration and customization options for its control interface, including joysticks, buttons, toggles, and sliders. Additionally, the 10" Toughpad FZ-G2 Panasonic tablet, which serves as the controller's display and interface, is replaceable, allowing for easy maintenance and upgrades.

6. Control Interface: A Closer Look

1. Front Side

At the front of the 10" Maxi Controller Pro, we have customizable joysticks, buttons, toggles, and sliders for precise control. The 10" Toughpad FZ-G2 Panasonic tablet serves as the central display and interface.

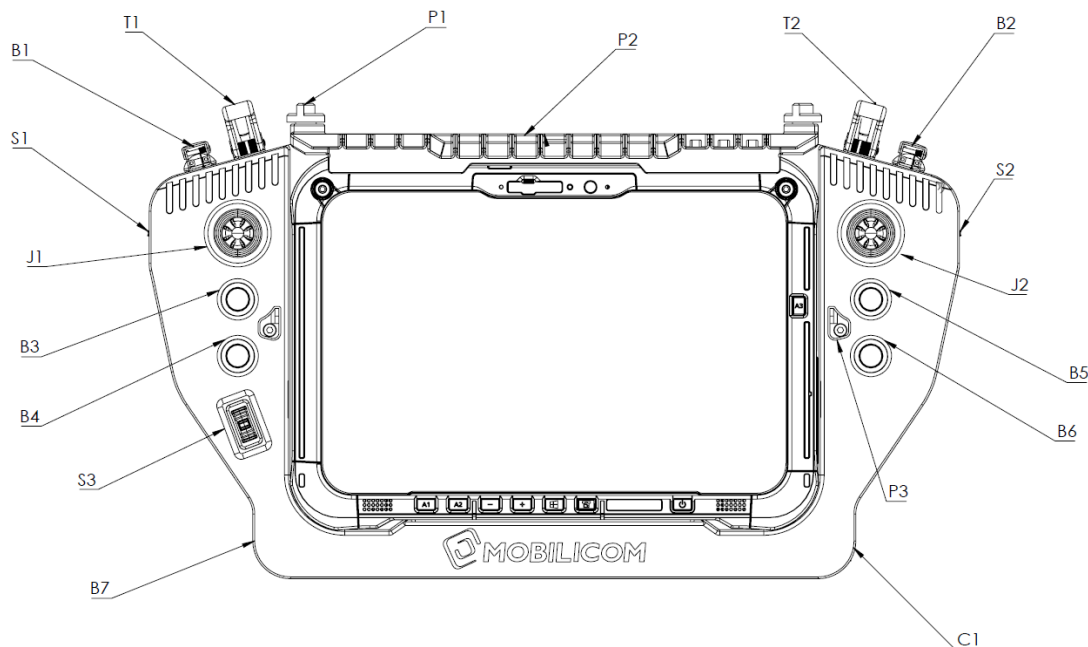


Figure 2: 10" Maxi Controller Pro Buttons, Joysticks, toggles, and sliders

2. Back Side

The back side of the 10" Maxi Controller Pro features the Controller Serial Number label and a SIM card door for SIM card installation (applicable for SkyHopper COMBO models). Additionally, it is equipped with Standard VESA 200x100 mounting holes, providing versatility for a range of mounting options including tripods, body straps, and more.

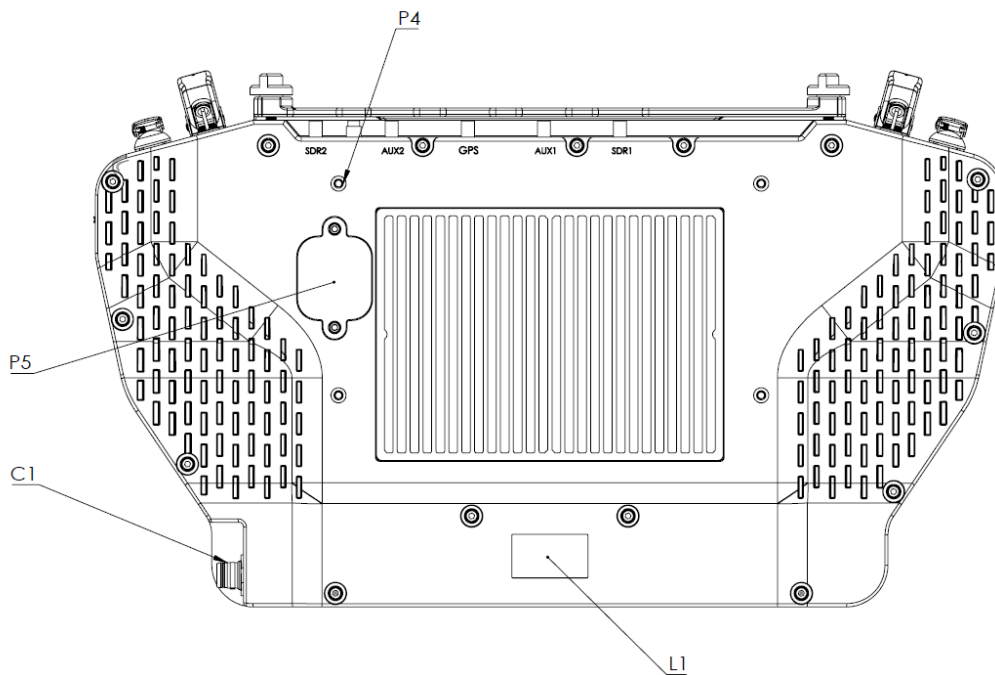


Figure 3: Back Side

3. Mapping Table

#	type
T1, T2	Toggle switch (Optional for safety cap) T1 & T2 working state is detailed in Table 4
B1, B2, B3, B4, B5, B6	Momentary push button
B7	Stady state On/Off push button. On/Off power for SkyHopper Pro
S1, S2, S3	Slider
J1, J2	Joystick
P1	Loker Screw
P2	Mounting Plate
P3	Body streps holder
P4	VESA 200x100 mounting holes
P5	SIM card cover
L1	S/N label

Table 2: Mapping Table

7. Safety Precautions

To ensure optimal performance and functionality of the 10" Maxi Controller Pro, please follow the following mandatory operational instructions.

- Avoid replacing the 10" Toughpad FZ-G2 Panasonic while DC power is connected to the 10" Maxi Controller Pro.
- Always power off the tablet before inserting or removing it from the 10" Maxi Controller Pro.
- Ensure the tablet is inserted parallel to the casing for proper alignment.
- Use hand tightening only when closing locker knobs; avoid using tools for leverage and ensure both knobs are evenly tightened.
- Ensure that antennas are connected before powering up the SDR unit. Failure to do so may result in irreparable damage to the unit.

8. 19-Pins Harness Connector

The 10" Maxi Controller Pro features a 19-pin harness connector located on its right side, designed for connecting an external harness. This harness facilitates various functions, including power input for charging and extended usage, 10/100 Ethernet connectivity, and USB 2.0 interfaces. Please see Appendix E for detailed pinout.

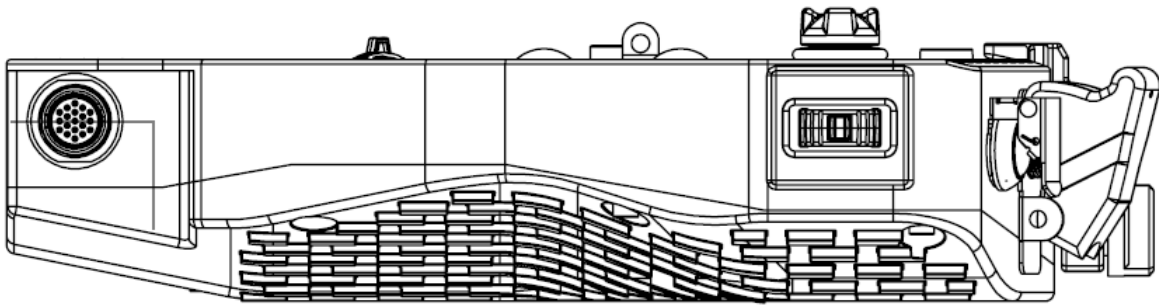


Figure 4: 19 Pin Harness Connector

9. Connecting the Antennas

Connecting the antennas to the 10" Maxi Controller Pro is a crucial step for setting up the device. Here are some additional details to keep in mind when connecting antennas:

1. **Antenna Placement:** Antennas should be positioned as high and clear as possible to minimize obstructions and maximize signal strength.
2. **Antenna Selection:** Choose antennas based on the specific application and requirements. Consider using directional or omnidirectional antennas, depending on the scenario.
3. **Cable Selection:** Select cables carefully, considering factors like attenuation and flexibility. Lower attenuation cables offer better signal strength, while more flexible cables ease installation.
4. **Antenna Orientation:** The orientation of the antenna influences signal strength and quality. Verify that the antenna positioning is in accordance with the antenna type and its patterns.

For SkyHopper COMBO usage, also connect the LTE antennas.

Connecting Power:

Caution: Powering up an SDR unit without connecting the antennas may result in irreparable damage to the unit.

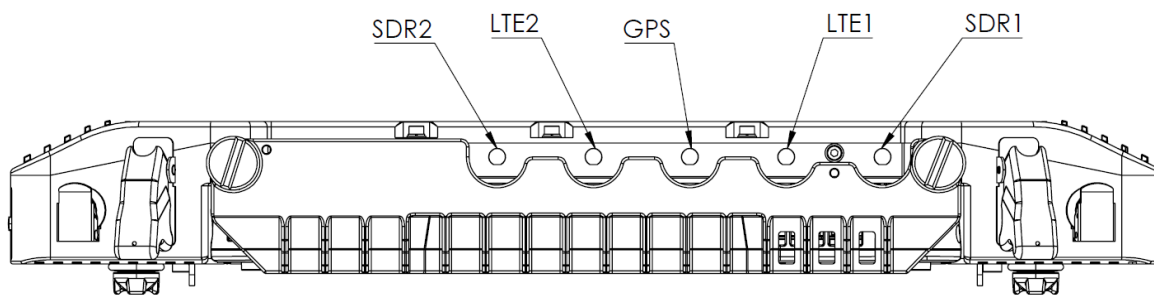


Figure 5: Antenna SMA connectors.

10. Panasonic FZ-G2 Toughpad

10" Maxi Controller Pro includes the Panasonic FZ-G2 Toughpad. In order to turn the tablet ON, press and hold the Power Switch  until the Power Indicator  lights up.



Figure 6: Panasonic FZ-G2 Toughpad

For full specifications and additional information please access:

<https://eu.connect.panasonic.com/fi/en/products/toughbook/toughbook-g2-standard>

11. Joystick Calibration

To calibrate the joysticks, follow these steps.

1. Press the Windows Button and type “USB game” to access the USB game controller.
2. Select the APEM USB Kit and press Properties.
3. Click OK in the popup window.
4. Navigate to Settings and select Calibrate.

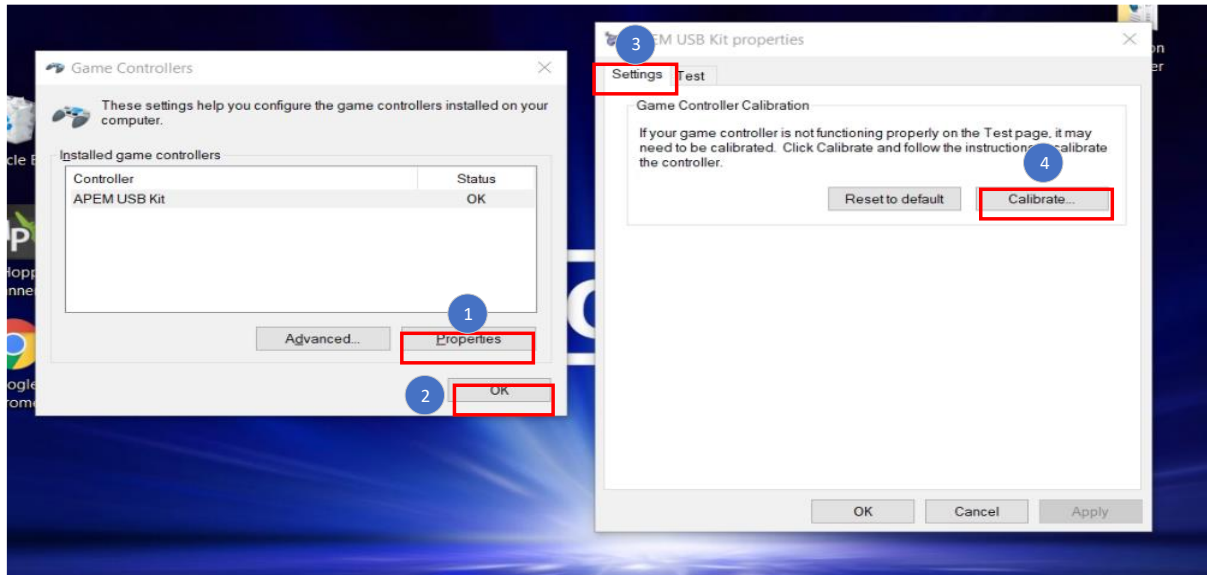


Figure 7: Joystick Calibration

1. Press "Next"
2. Calibrate the right joystick by rotating it clockwise. Ignore erratic behavior at this stage.

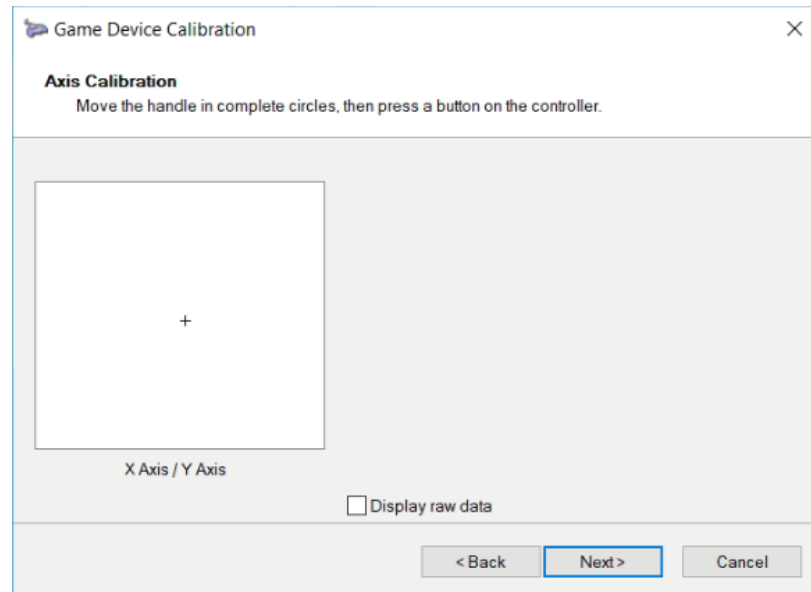


Figure 8: Right joystick Calibration

3. Press "Next"
4. Verify the center point and press next.
5. Move the Z-axis button up and down 3 times and then press "Next".

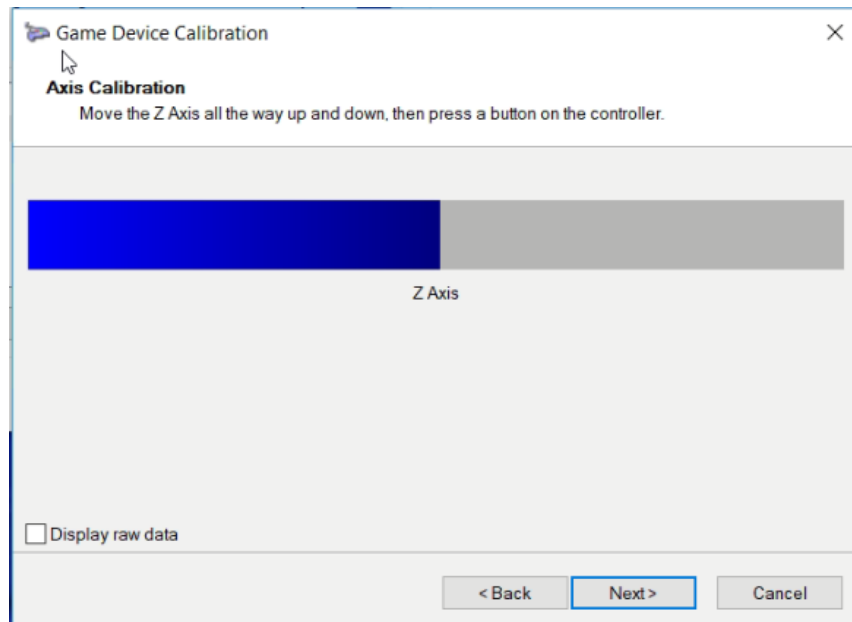


Figure 9: Z-Axis Calibration

6. Move the X-Rotation button right and left 3 times and press "Next"

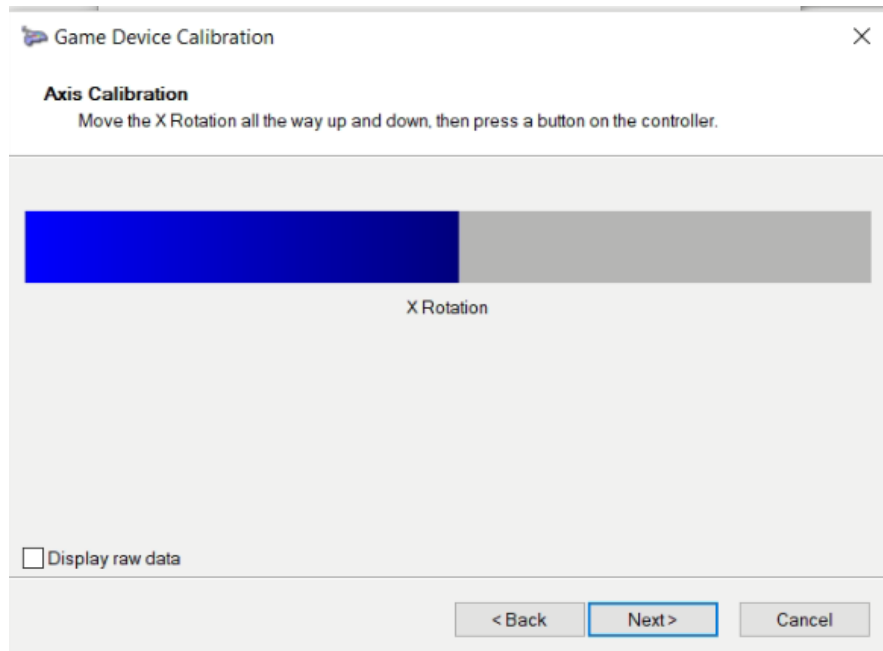


Figure 10: X-Rotation Calibration

7. Move the Y-Rotation button up and down 3 times and press "Next".

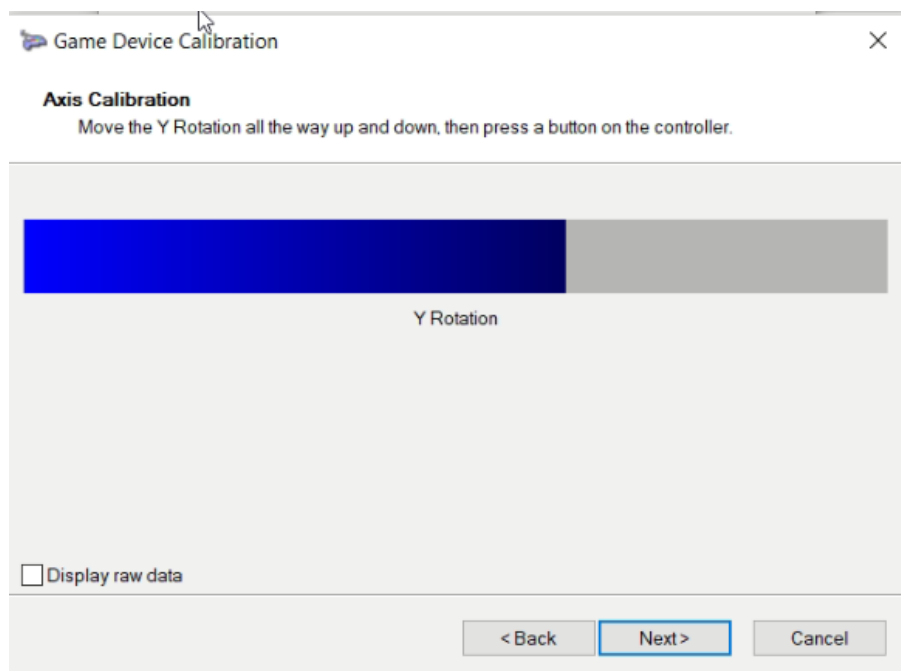


Figure 11: Y-Rotation Calibration

8. Move the Z-Rotation button left and right 3 times and press "Next".

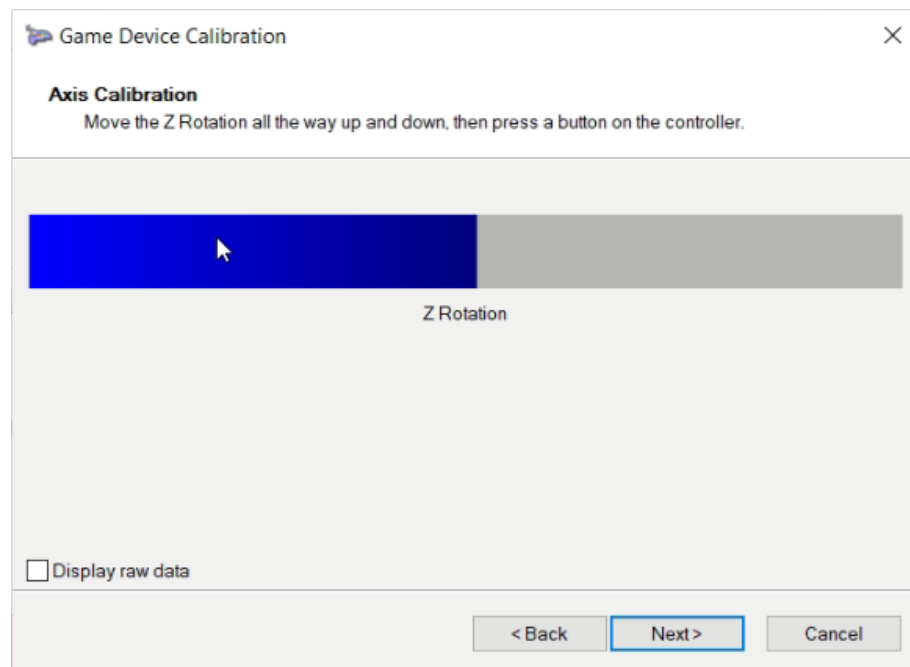


Figure 12: Z-Rotation Calibration

9. Press "Next"(Don't move any button).

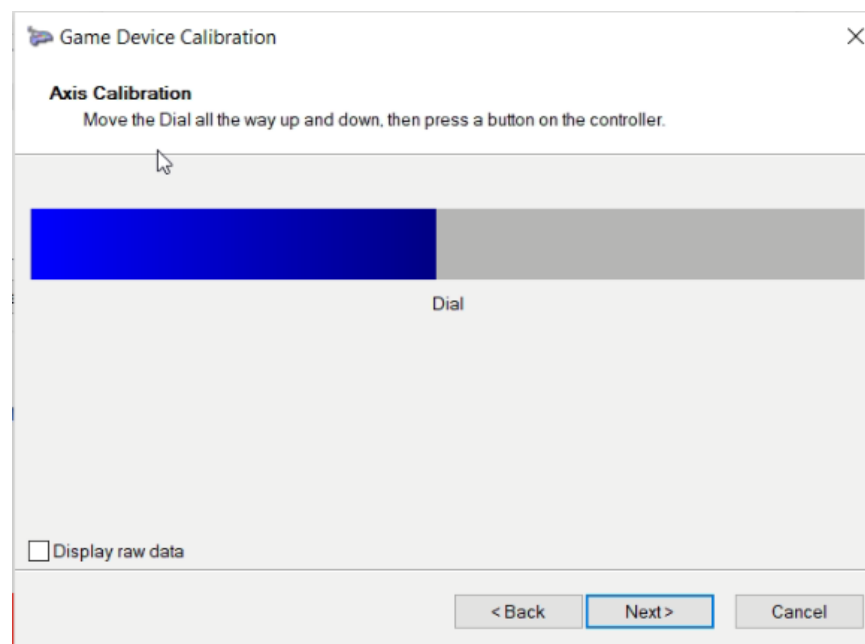


Figure 13: Dial Calibration

10. Move the slider up and down 3 times and press "Next".

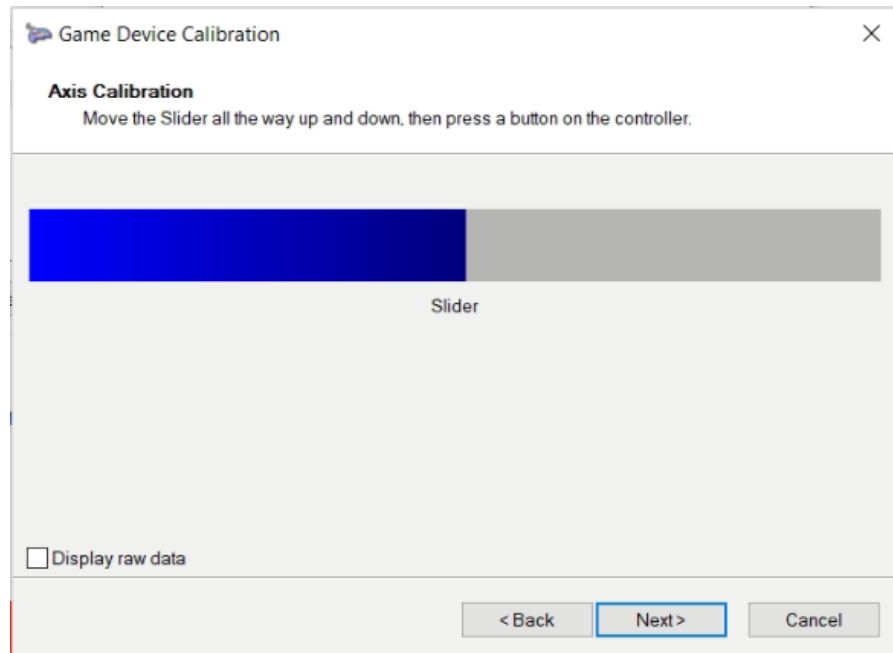


Figure 14: Slider Calibration

11. Press Finish

Press the buttons according to chart and View output for each scenario:

Press on	Expected Output
Button1	Light on output 7
Button 2	Light on output 15
Button 3	Light on output 1
Button 4	Light on output 2
Button 5	Light on output 3
Button 6	Light on output 4
Button 7 – No output on game controller	Stady state on off push button Power On/Off for SkyHopper Pro
Toggle1=Up, Toggle2=Up/Down	Light on output 8
Toggle1=Down, Toggle2=Up	Light on output 16
Toggle1=Down, Toggle2=Down	Light on output 24

Table 3: Buttons Testing

13. SDR Unit Power Switch

The 10" Maxi Controller Pro features a dedicated Power Switch for the internal SDR (Software Defined Radio) unit located on the left side. This switch allows RF communications to be turned on and off while still enabling functionality with the tablet.

1. Turning On the SDR Unit: Press the Power Switch and ensure it remains pressed, with the LED light indicator turning on.
2. After pressing the button, access the unit's Web-GUI to verify that the unit is ON.
3. Turning Off the SDR: Press the Power Switch again and ensure that the LED light indicator turns off, indicating that the unit is OFF.
4. The internal SDR units can include either Mobilicom's SkyHopper PRO SkyHopper COMBO, or MCU-30.

Warning: Before powering up the SkyHopper PRO unit, ensure that antennas are connected to RF1 and RF2 connectors to prevent irreparable damage to the unit.



Figure 16: Internal SDR Unit Power Switch

Appendix A – Inserting/Replacing the tablet

Warning: Do not replace the 10” Toughpad FZ-G2 Panasonic while DC power is connected to the 10” Maxi controller Pro.

Warning: Always turn off the Tablet before inserting or extracting it from the 10” Maxi Controller Pro.

Warning: Always insert the tablet in parallel to the casing.

Warning: Close locker knobs by hand only (do not use tools for leverage). Both knobs should be evenly tightened.

To **remove** the FZ-G2 Toughpad in the 10” Maxi Controller Pro, please follow the correct procedure:

1. Make sure the Toughpad tablet is powered off and disconnected from the external power supply.
2. Gently loose and remove both upper Tablet locking knobs by turning them counterclockwise.
3. Remove the top cover.
4. Gently remove the Toughpad tablet by pushing it upwards.



Figure 17: Top Cover and Tablet Locking Knobs

To **insert** the FZ-G2 Toughpad in the 10” Maxi Controller Pro, please follow the correct procedure:

1. Make sure the Toughpad tablet is powered off and disconnected from the external power supply.
2. Gently insert the FZ-G2 Toughpad in parallel to the 10” Maxi Controller Pro slot, without angles, and ensure the proper connection of the tablet’s holders and pins.
3. Place the top cover aligned with the screw bolts on the top of 10” Maxi Controller Pro.
4. Gently insert the tablet locking knobs and turn (by hand only, without tools) them clockwise until they are firmly securing the top cover in place.

Appendix B - Troubleshooting

Symptom	Possible Cause	Action Needed
Tablet won't turn ON	Not connected to power source or battery is empty	Verify power cable properly connected to the 10" Maxi Controller Pro power connector. Verify that the battery of the Toughpad Tablet is charged. Replace the battery or check the outlet. Replace the power cable.
No link between SkyHopper PRO units	Configuration isn't the same on all the SkyHopper PRO units. PTP – both units set as Remote or both as Controller. The units aren't within reception range	Connect to each unit and verify they have the same configuration. Verify one unit is set as a Remote and the other as a Controller. Verify that TX power is set to the maximum. Verify the antennas fit the Frequency used (check the antenna spec).
Data with high delay and disconnecting data	Trying to send a higher amount of data than the allocated bandwidth sent to the unit	Verify that the data bandwidth sent to the unit from the connected device(s) does not exceed the allocated bandwidth (in its configuration). Decrease the data received by the data source or change the allocated bandwidth via the Web GUI.

Table 4: Troubleshooting

Appendix C - Mechanical Drawings

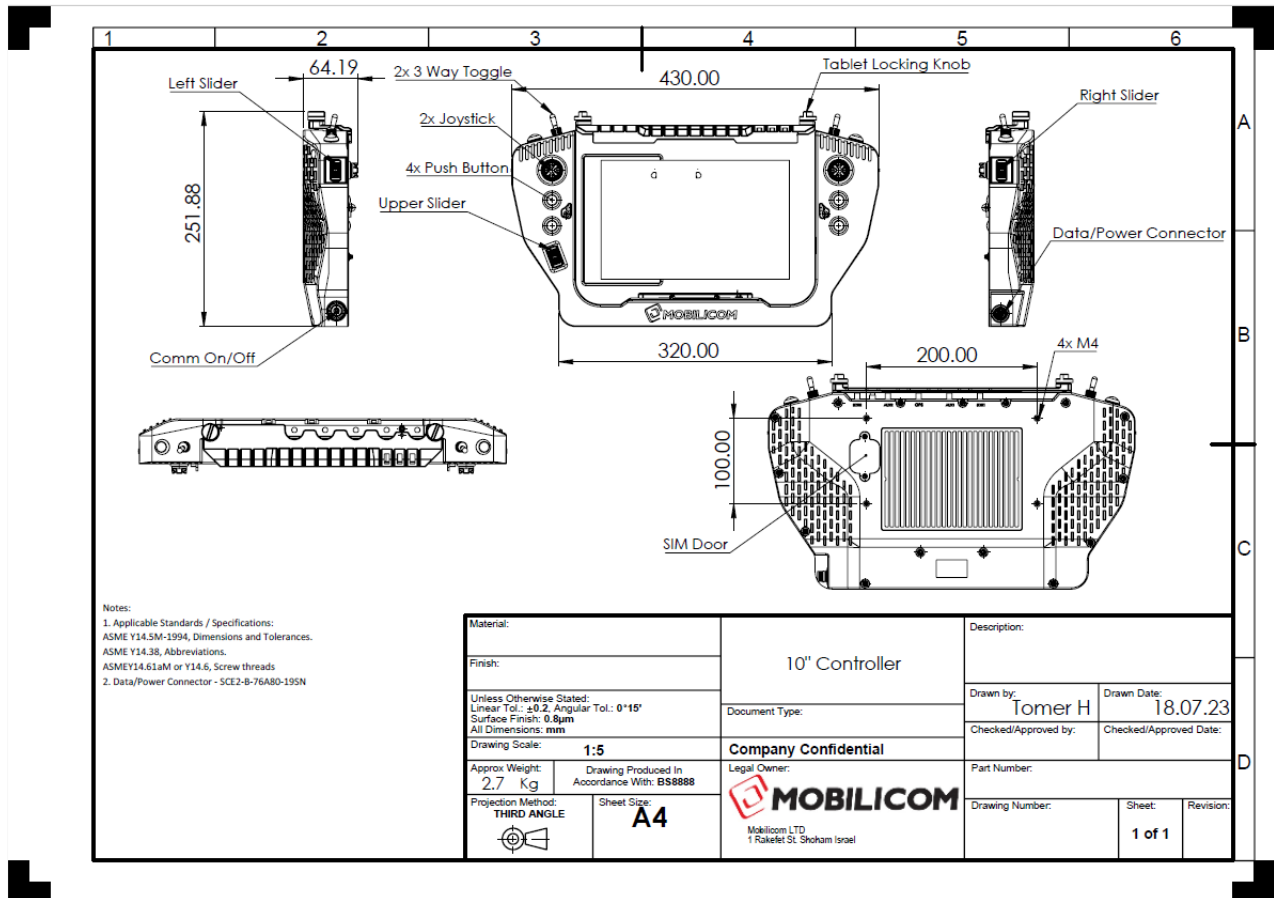


Figure 18: 10" Maxi Controller Pro Mechanical Drawing

Appendix D – Harness Schematics

The 10" Maxi Controller Pro has Power, ETH, and USB interfaces (Connector 19 Pins). The connector part number is SCE2-B-76A08-19SN Amphenol. The schematics are as follows:

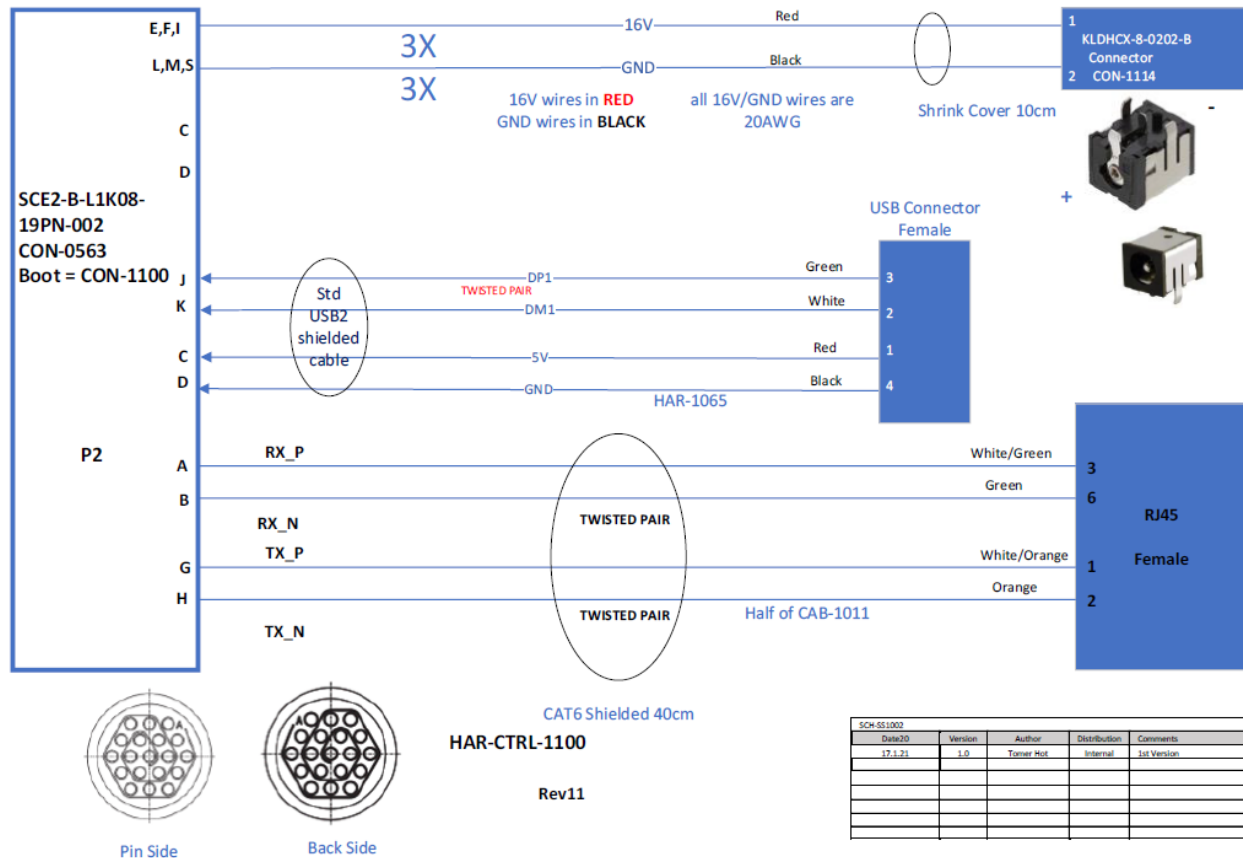


Figure 19: Harness Schematics

For technical support for 10" Maxi Controller Pro, please contact:

support@mobilicom.com